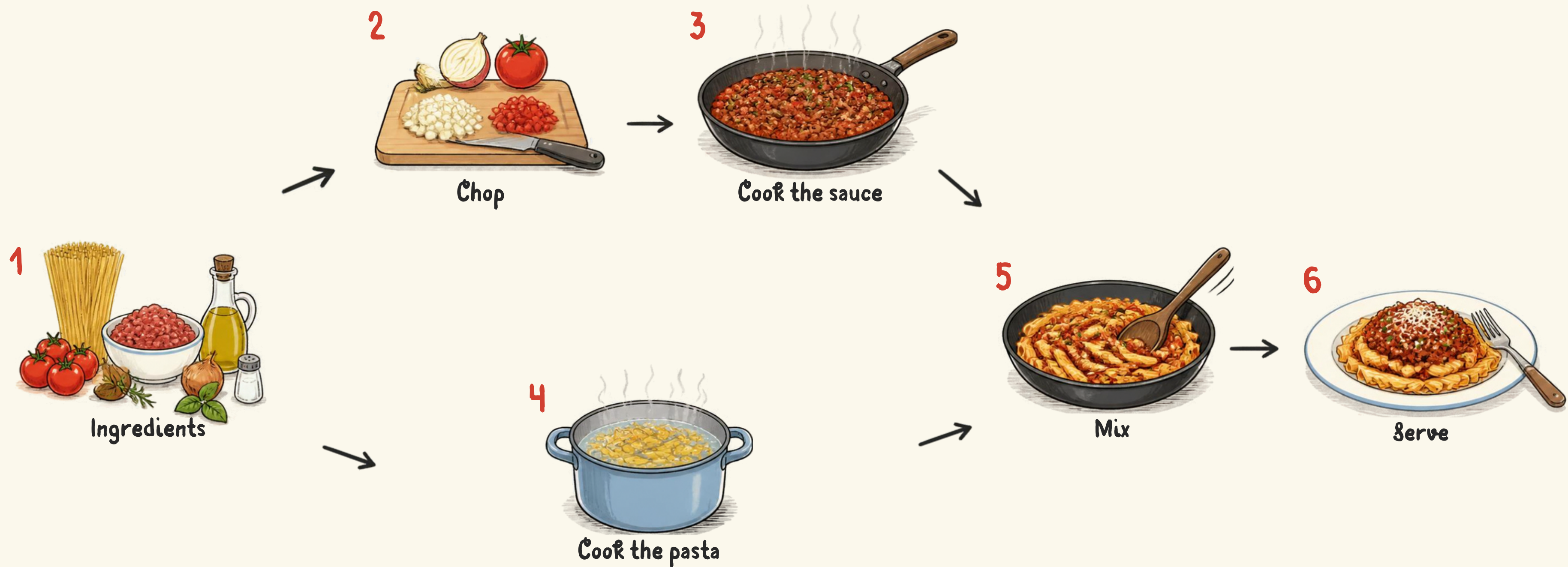


# Introduction to programming and vibe coding

- **Programming Concepts Explained Through a Recipe**
- **Hands-on Workshop**
  - **Introduction to Python**
  - **Introduction to Vibe Coding**
  - **Coding with an AI Agent: A Live Demo**

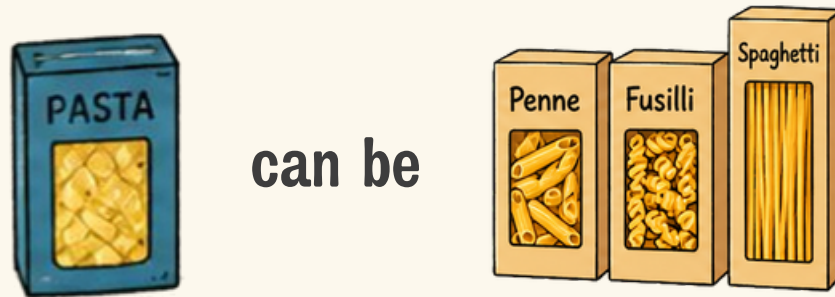
# Pasta with Bolognese sauce



# A recipe, seen as a program: Variable

## 1. Cook the pasta

Variables



Let's choose fusilli for our pasta



then

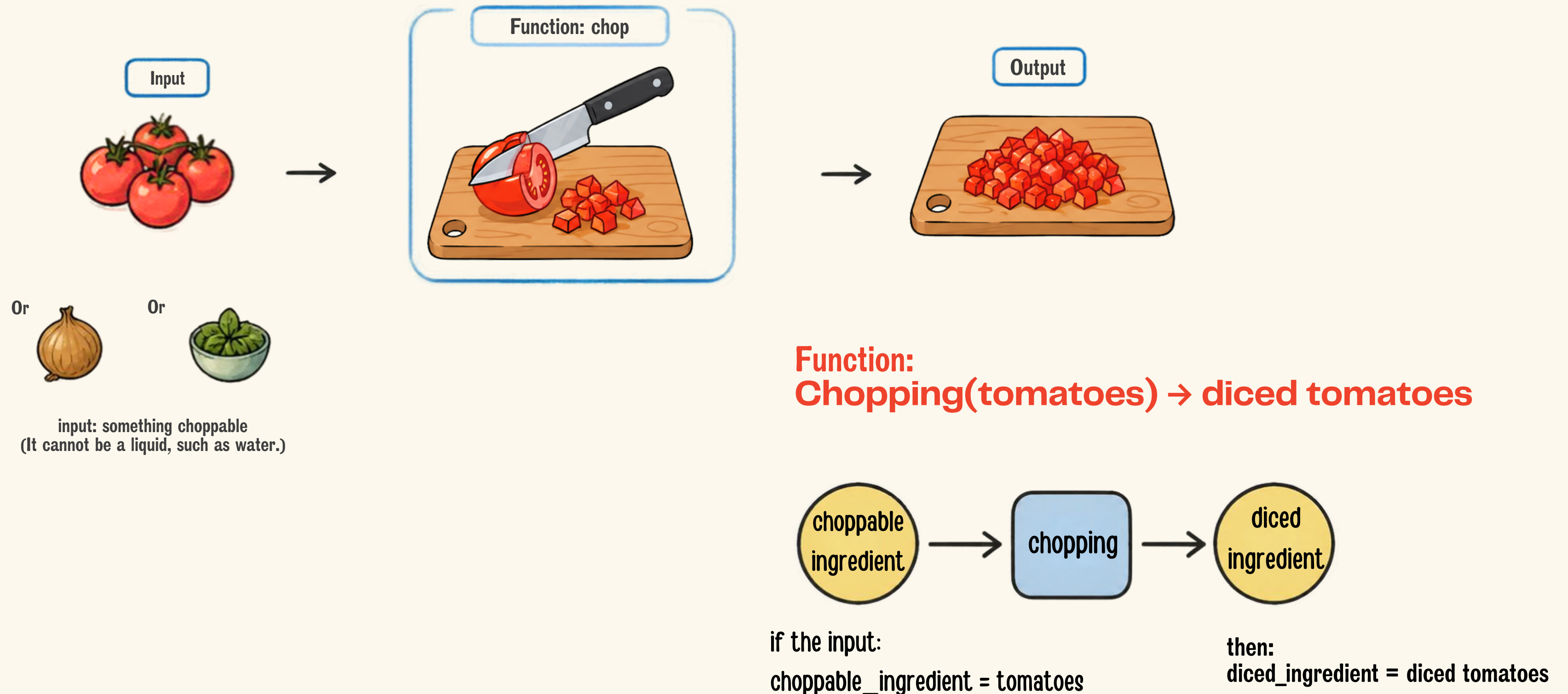


then

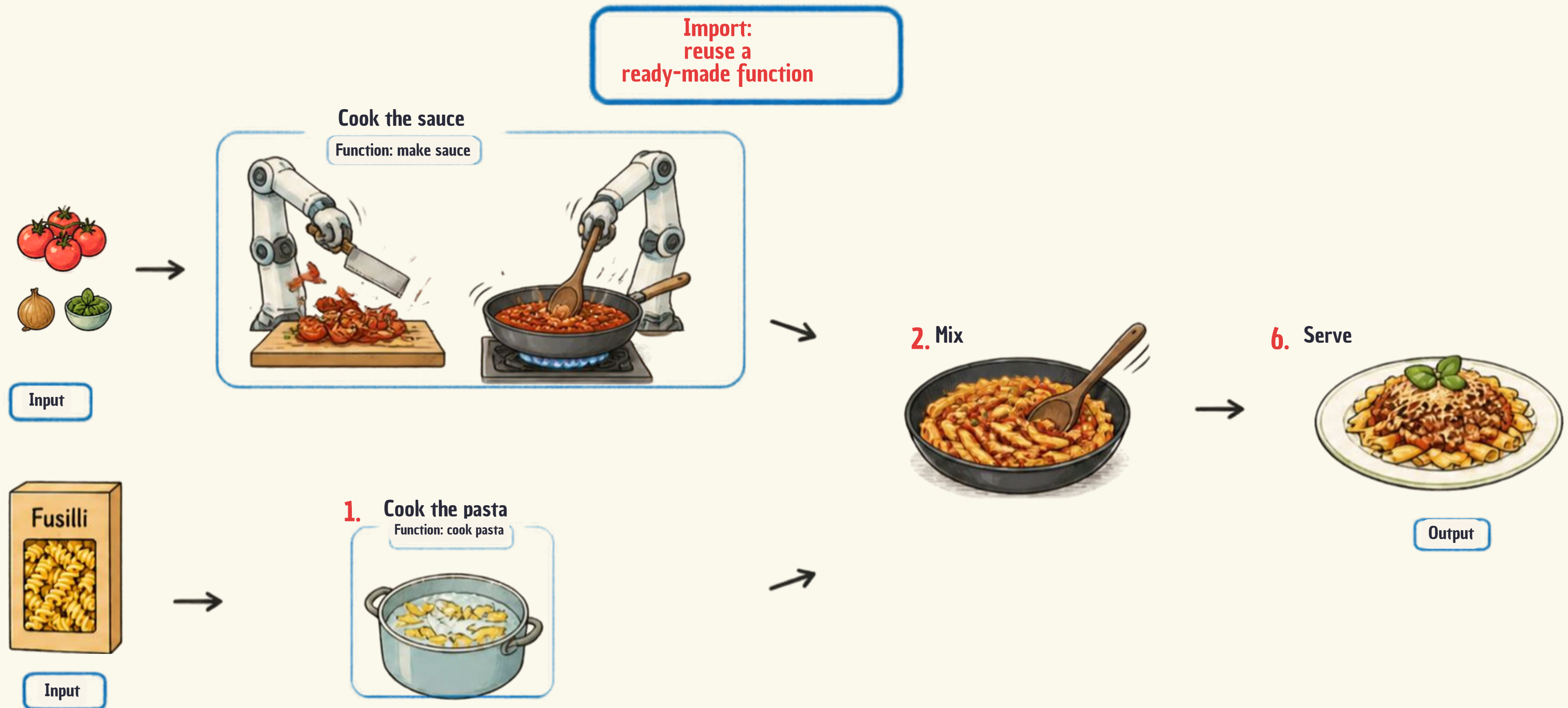


# A recipe, seen as a program: Function

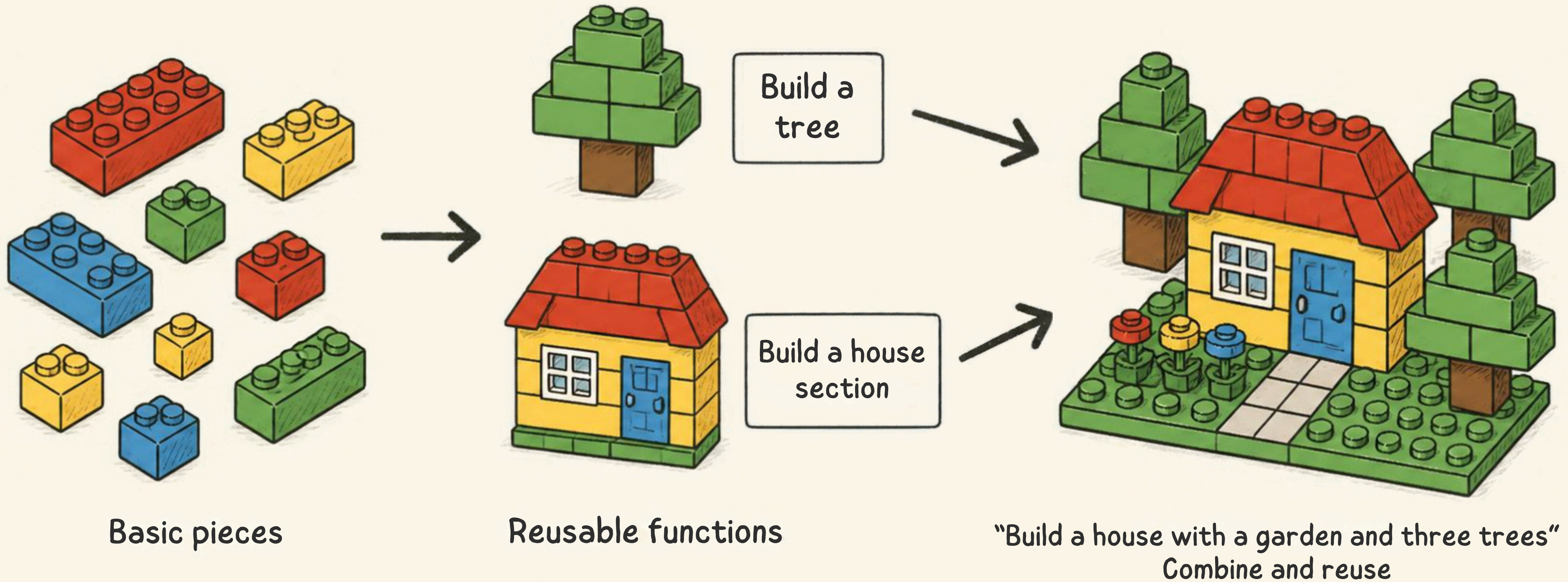
## 2. Chopping tomatoes



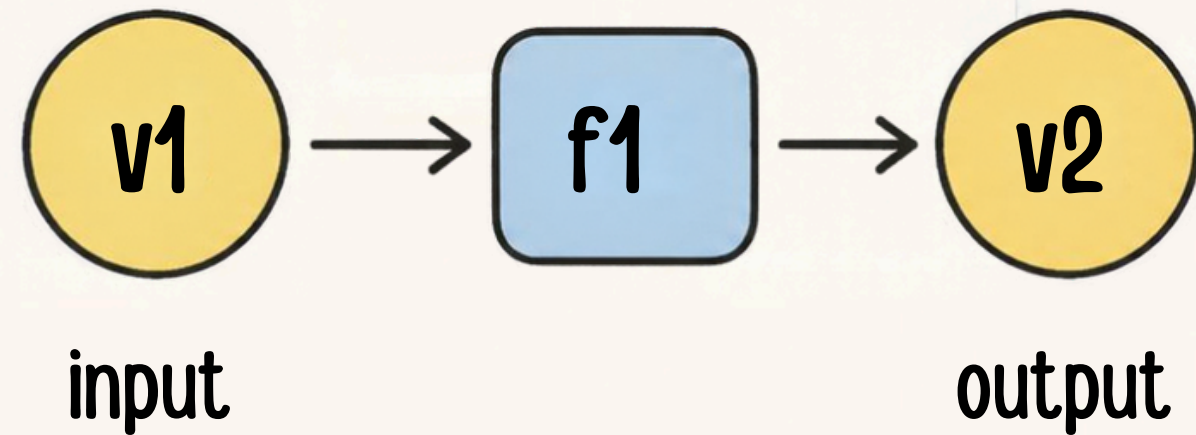
# A recipe, seen as a program: Import



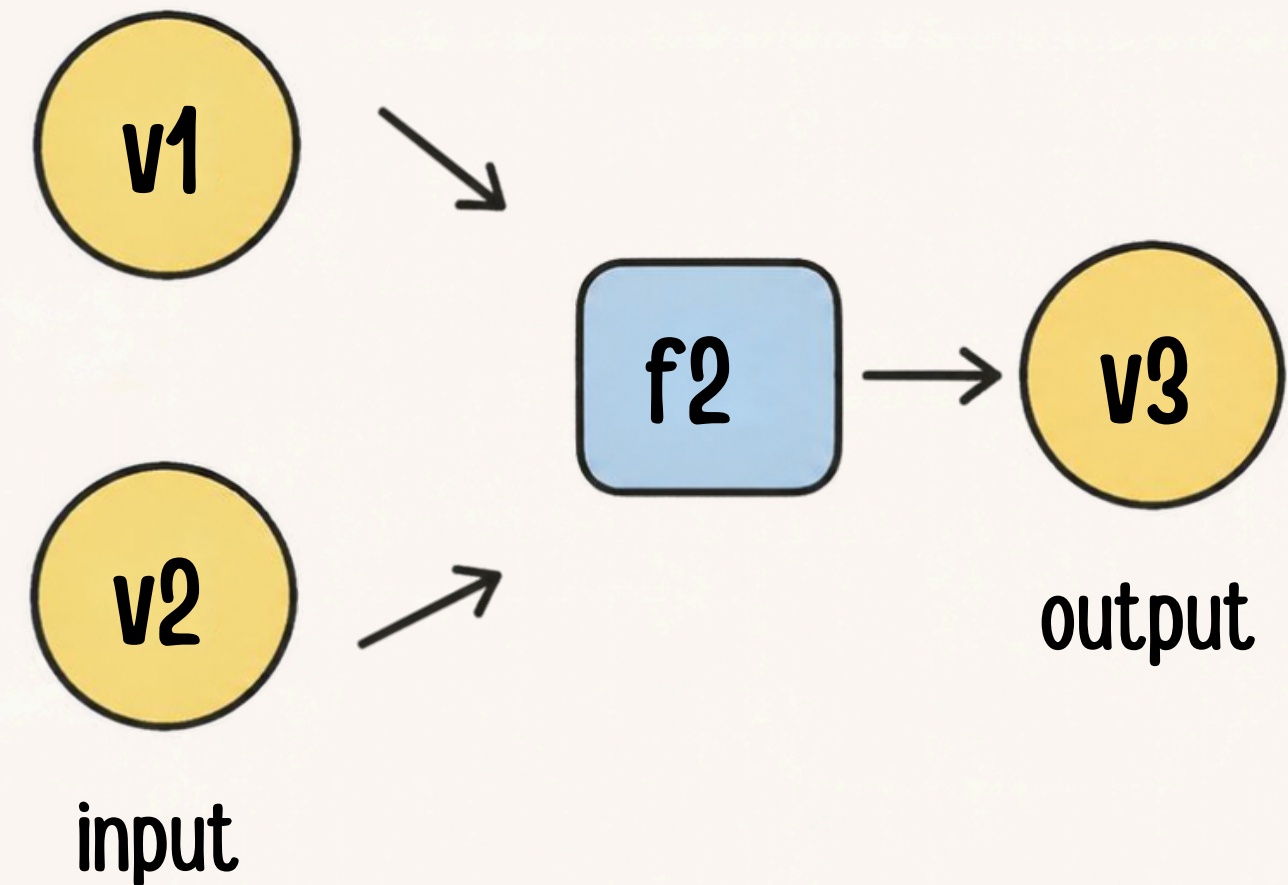
# Build once, reuse



# From recipes to programming diagrams

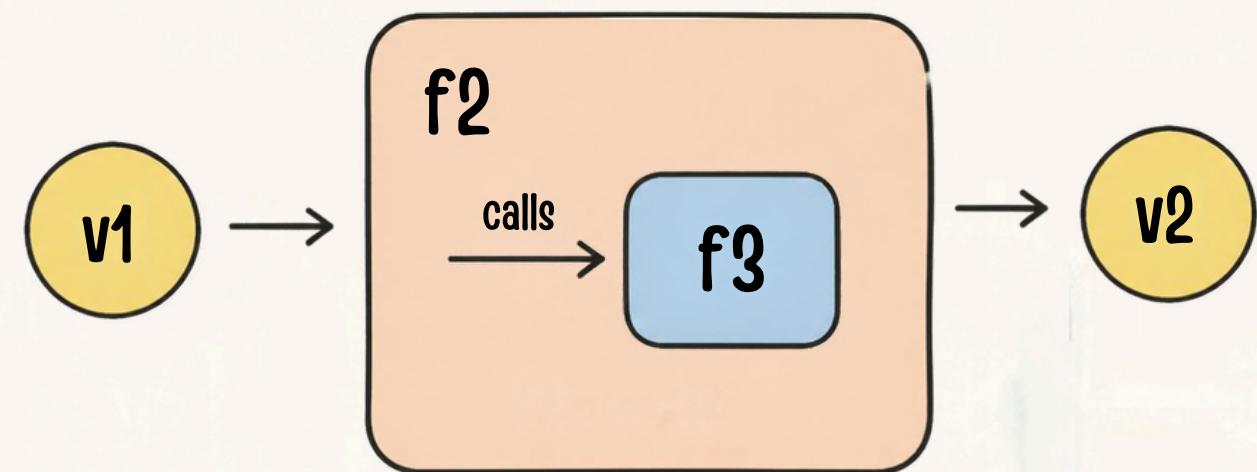


raw pasta → boil → cooked pasta  
25 °C → convert → 77 °F



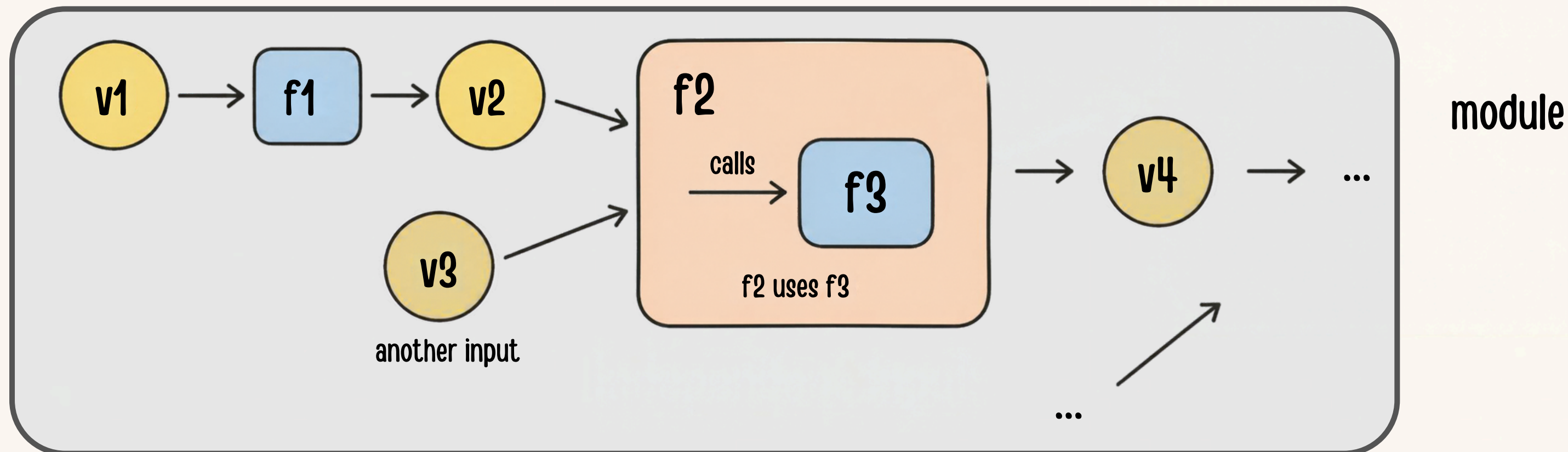
tomatoes → cook → Bolognese sauce  
minced beef ↗

# Functions can work together (1)

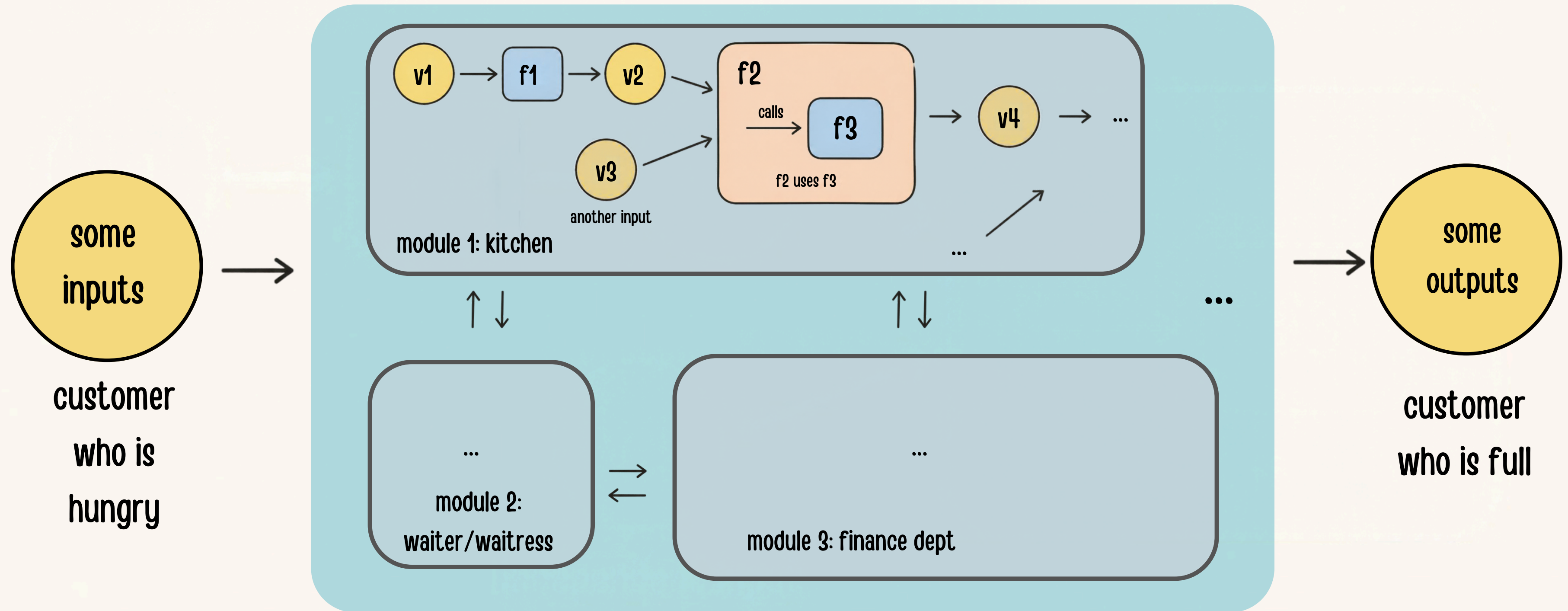


**f2 uses f3:**

The chef (**f2**) prepares a dish (**v2**) with help from an assistant (**f3**), who chops the ingredients.



# Functions can work together (2)



combine modules together to make more powerful software

# Hands-on workshop:

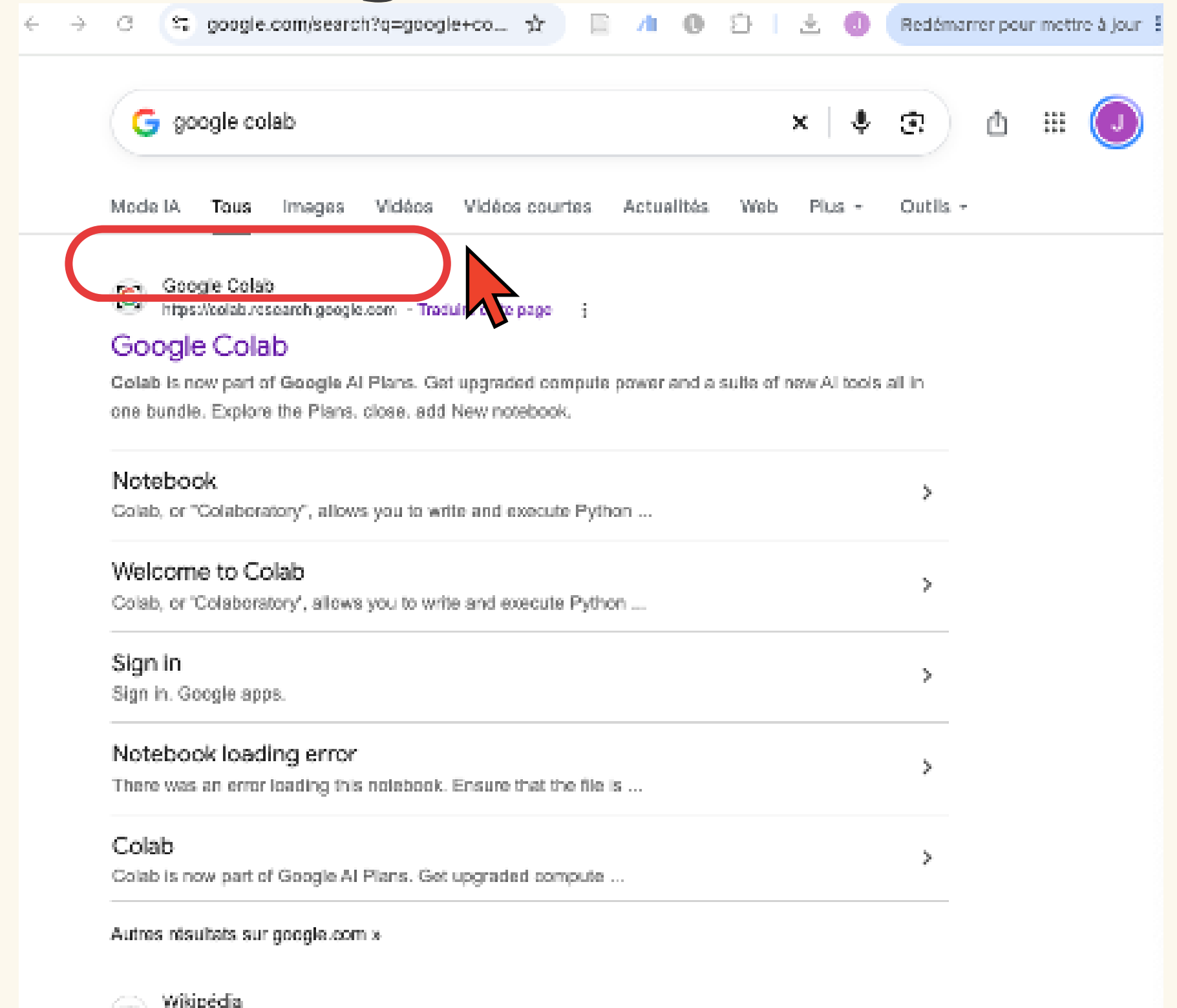
## First step: open the notebook in Google Colab

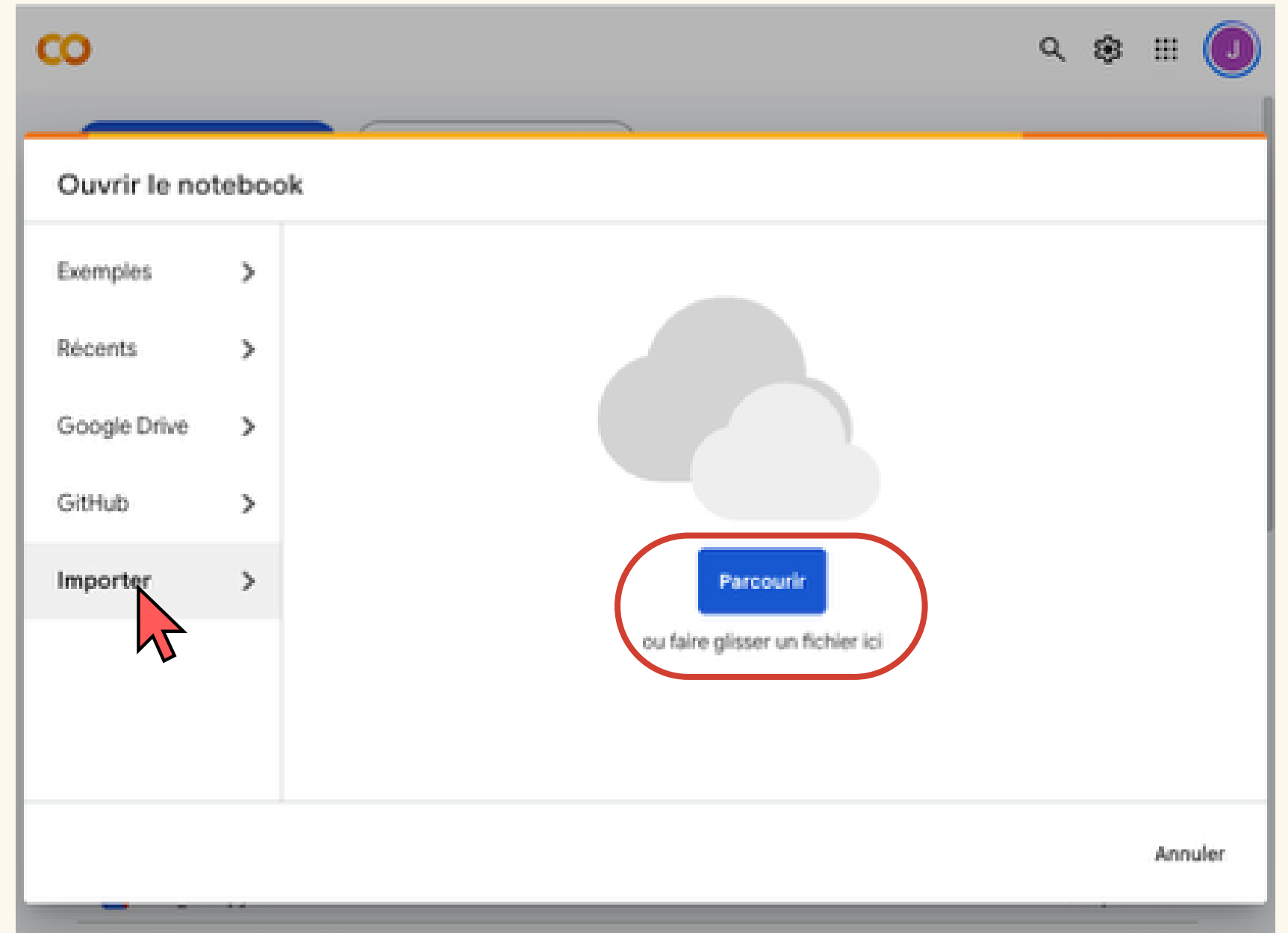
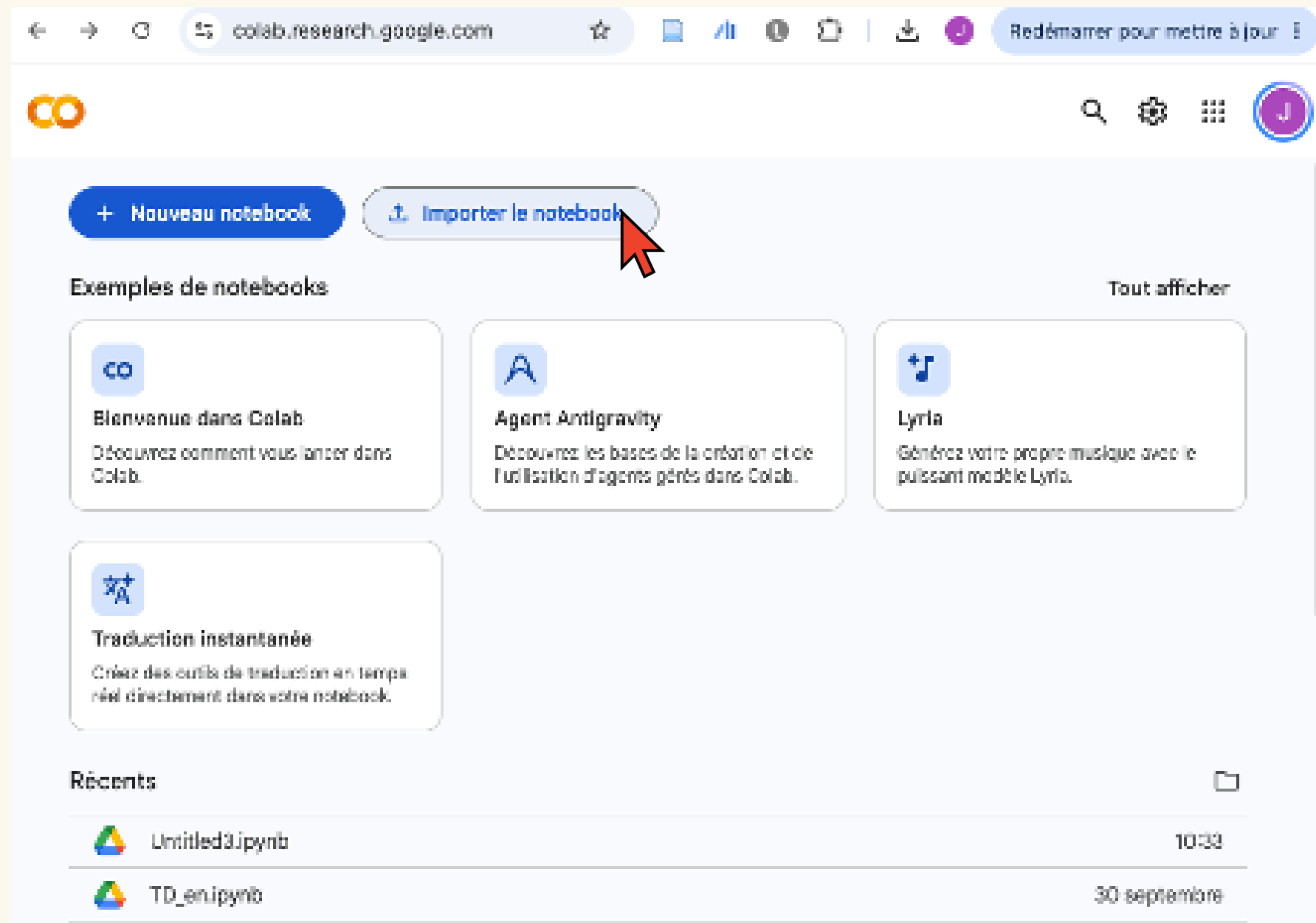
We will use Google Colab to run our Python code in this workshop.

Google Colab is a website where you can write and run Python code in your browser without installing anything on your computer.

We will use a type of document called a notebook for today's exercise.

A notebook is an interactive document that combines text, code, and results, allowing you to run code in separate blocks called cells.





The screenshot shows a Google Colab notebook interface. The browser address bar displays the URL: `colab.research.google.com/drive/1f90tYHdi3e-PbKaiv3ud5bD...`. The notebook title is `TD_student.ipynb`. The top navigation bar includes a menu with options: `Fichier`, `Modifier`, `Affichage`, `Insérer`, `Exécution`, `Outils`, and `Aide`. On the right, there is a `Partager` button and a user profile icon. Below the navigation bar, a toolbar shows `Commandes`, `+ Code`, `+ Texte`, and `Tout exécuter`. On the far right of the toolbar, there are status indicators for `RAM` and `Disque`. The left sidebar contains icons for file management and execution. The main content area is titled `Answers to common questions` and contains a bulleted list of instructions:

- To run a code cell, click the play button or press `Shift` + `Enter`.
  - I recommend running the code blocks **one by one, without skipping any**, because later blocks may depend on the results of earlier ones.
- To search for text in the file, press `Ctrl` + `F` on Windows or `Command` + `F` on macOS.
- A script contains two types of text:
  - **Comments:** in Python, comments begin with `#`. Use everyday language to explain the purpose of the code or the logic of an algorithm. Comments are for humans; Python Interpreter ignores them.
    - In this notebook, comments explain the code and describe tasks.
  - **Code:** instructions that must follow the programming language's syntax.
    - **Rule No. 1 in Python: indentation matters. Adding spaces at the beginning of a line can show that it belongs to the instruction above. Think of it as a sub-step: it tells Python, "this is part of that instruction."**
- Ask your LLM assistant for help when you encounter:
  - A line of code you do not understand.

You should see a page like this.

Congratulations! You are ready for the workshop!